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Understanding the Smith chart

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Understanding the Smith chart

Author: Paul Denisowski, Product Management Engineer

The **Smith chart** is an essential tool in **radio frequency (RF) engineering** and **circuit design**. Named after its inventor, Phillip Hagar Smith, the Smith chart was first introduced in Electronics magazine in January 1939. Smith later authored a comprehensive 200-page book detailing the numerous applications of his invention. The Smith chart was originally created to simplify mathematical calculations and allow problems to be solved graphically using a compass, ruler and pencil. However, it is still relevant today due to its utility in **visualizing complex impedances**, especially as a function of frequency. It is widely used to **tune and verify matching networks**.

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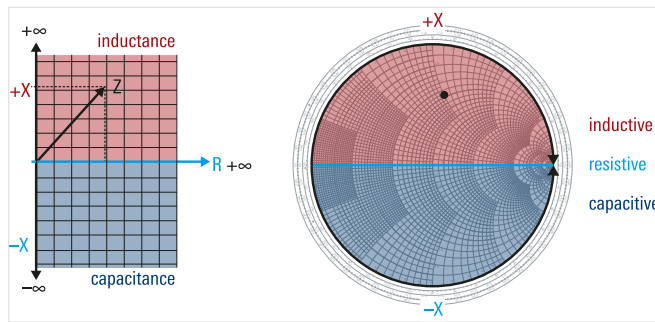
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Smith chart basics

The **Smith chart** is primarily used when making **one-port measurements**, particularly to **visualize reflection coefficients**. It represents the load impedance, Z_L , relative to the source impedance, Z_0 . **Complex impedance** values can be plotted as individual points or as lines that depict impedance over a range of frequencies.

In **Cartesian coordinates**, the complex impedance is represented by a resistive part (R) and a reactive part (X). Traditional Cartesian plotting has limitations due to its infinite range for both impedance and resistance. The Smith chart resolves this by effectively **bending the right half of the Cartesian coordinate plane**, bringing the positive and negative reactance axes around to meet the resistance axis. This results in a circular chart

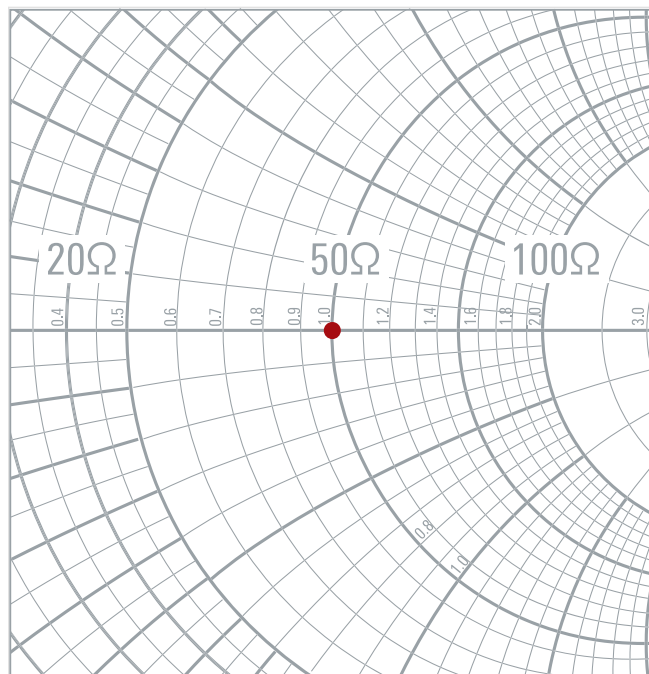
where the upper half represents the **inductive region** and the lower half represents the capacitive region, with a purely **resistive axis** separating the two.



Cartesian coordinates to Smith chart

Smith chart impedance matching

Let's start by exploring the center of the Smith chart, also known as the **prime center**. This point corresponds to Z_0 . In most RF systems, Z_0 is a **purely resistive 50-Ohm load**. The Smith chart normalizes this source impedance to 1. In other words, the center of the chart, marked as 1.0, represents a purely resistive 50-Ohm load. Moving along the resistive axis to 2.0, for example, would indicate a pure resistance of 100 Ohms (2 times 50), while moving to 0.4 would correspond to a resistance of 20 Ohms (0.4 times 50). All values on the Smith chart are normalized in this way, allowing it to be used in systems with different impedances, such as 75 or 60 Ohms.



Prime center of the Smith chart

For optimal power transfer and to minimize reflected power, **Z_L should closely match Z_0** . In other words, a key goal in **impedance matching** is to make Z_L as close to the center of the Smith chart as possible.

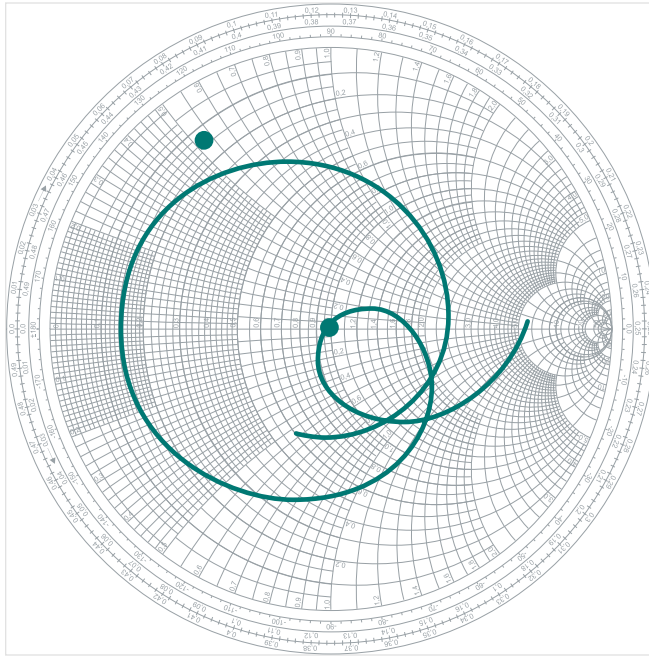
Measured values of Z_L are plotted on the Smith chart, where the Z_0 is always at the center.

The closer the measured Z_L values are to the center, the better the impedance match.

A perfect match is achieved when the plotted value is at the center of the chart.

The farther away a point is from the center, the higher the degree of mismatch.

If a trace of Z_L is plotted as a function of frequency, the load is resonant at the frequency where the trace moves through or near the center of the Smith chart.

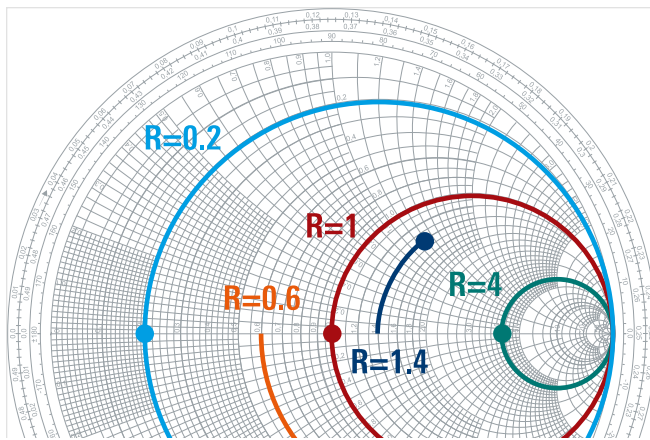


A device is resonant at the frequency where the trace moves through the center

Resistance and reactance on the Smith chart

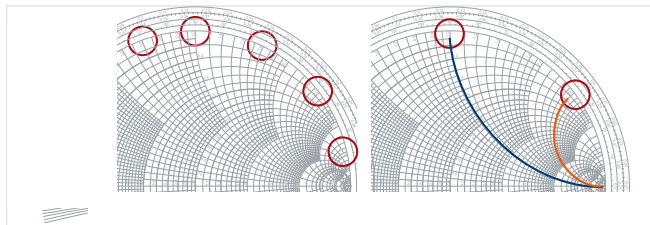
The **resistance axis** is the only straight line on the Smith chart. The normalized, purely resistive source impedance is represented by the "1" in the center, corresponding to a **voltage standing wave ratio (VSWR)** of 1:1. Moving left along the axis, resistance decreases until it reaches the edge of the circle, representing zero resistance or a **short circuit**. Moving right, resistance increases towards infinity, representing an **open circuit**. Points on this resistance axis have pure resistance with no reactive component, while any point along the edge of the Smith chart represents a situation where VSWR is infinite and 100% of the power is reflected.

Most loads have both **resistive** and **reactive** components, so their impedance values will not lie directly on the resistance axis. Instead, the resistive part of a complex impedance will be found along a **resistance circle**. For example, the circle passing through the "1" on the resistance axis represents a normalized resistance of 1.0, meaning every point on this circle has a normalized resistive part equal to 1. Similarly, a circle passing through the point "0.2" on the resistance axis represents a normalized resistance of 0.2 at every point along the circle. To determine the resistive part of any complex impedance on the Smith chart, follow the corresponding resistance circle to where it intersects the horizontal resistance axis.



Normalized resistances are shown as resistance circles

The reactance of an impedance is also represented on the Smith chart. As previously mentioned, the **reactance axis**, which would be vertical in a Cartesian coordinate system, is curved around the circumference of the Smith chart. Values of normalized reactance are indicated along the chart's circumference, increasing from left to right. Similar to resistance circles, there are **reactance curves** that indicate constant normalized reactance values. Every point on a given reactance curve has the same reactive, or imaginary, part. The upper half of the Smith chart represents positive (inductive) reactance values, while the lower half represents negative (capacitive) reactance values.



Reactance axis and reactance curves

Plotting and interpreting complex impedances

With an understanding of resistance circles and reactance curves, plotting or interpreting complex impedances on the Smith chart becomes straightforward.

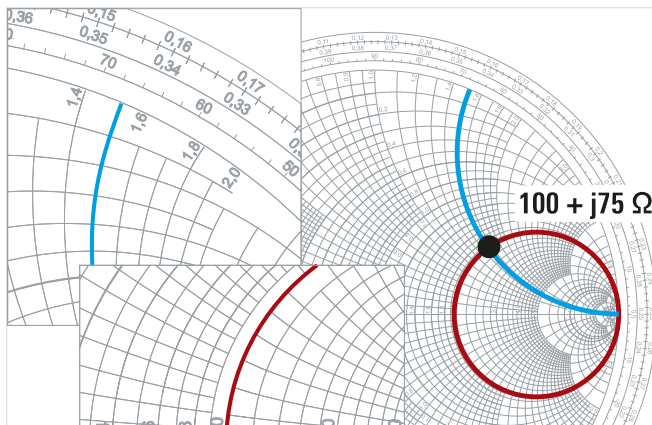
Let's walk through the steps of plotting an impedance of $100 + j75$.

Normalize the impedance by dividing both the real and imaginary parts by Z_0 , assumed to be 50 Ohms in this case. The normalized impedance is $2 + j1.5$.

Plot the resistance circle, which passes through the point 2 on the resistance axis.

Plot the reactance curve, which intersects the circular impedance axis at 1.5.

The point **where the resistance circle and reactance curve intersect** represents the impedance.



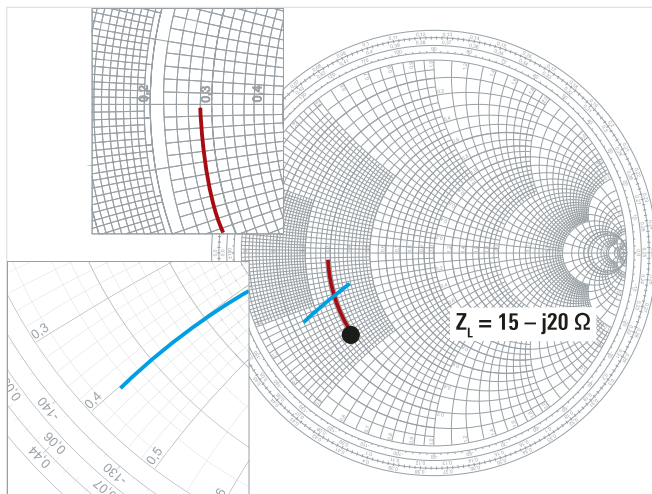
Impedance is where the red (resistance circle) and blue (reactance curve) lines intersect

You can reverse this procedure to determine a complex impedance from a Smith chart.

Identify the resistance circle the point lies on and the value at which the circle passes through the resistance axis.

Identify the reactance curve the point lies on and the value at which the curve passes through the circular reactance axis.

Multiply the normalized impedance by Z_0 to get the actual impedance.



Reading impedance from a Smith chart

Summary

The Smith chart displays complex impedances, which may be either individual points or lines showing values as a function of frequency. It enables many tasks, such as impedance matching, to be done graphically instead of algebraically.

A Smith chart consists of:

- Resistance and reactance axes
- Resistance circles and reactance curves

The values on a Smith chart are always normalized to the system (source) impedance.

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Address

1 Hines Road, Suite 100
Kanata (Ontario), K2K 3C7
Canada

Contact

Phone (+1) 613 592 8000
Toll free (+1) 877 438 2880
sales.rsc@rohde-schwarz.com

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